9.15 Solution: (a) From Prob. 6.21, we can see that the lowest non-vanishing moments are magnetic dipole and electric quadrupole. For a single dipole, its magnetic moment is

$$\frac{1}{2}\mathbf{p}\times\mathbf{v} = \frac{1}{2}\mathbf{p}\times(\boldsymbol{\omega}\times\mathbf{r}) = -\frac{1}{2}\omega pa\hat{\mathbf{r}} = -\frac{1}{2}\omega pa\left(\hat{\mathbf{x}}\cos\phi + \hat{\mathbf{y}}\sin\phi\right).$$

Therefore, for the two-dipole configuration as described in the problem, the magnetic dipole is

$$\begin{aligned}
\mathbf{m}(t) &= -\frac{1}{2}\omega pa\left[\left(\hat{\mathbf{x}}\cos(\omega t) + \hat{\mathbf{y}}\sin(\omega t)\right) - \left(\hat{\mathbf{x}}\cos(\omega t+\pi) + \hat{\mathbf{y}}\sin(\omega t+\pi)\right)\right] \\
&= -\omega pa\left(\hat{\mathbf{x}}\cos(\omega t) + \hat{\mathbf{y}}\sin(\omega t)\right) = \mathrm{Re}\left[\omega pa(-1,-i,0)e^{-i\omega t}\right].
\end{aligned}$$

For the quadrupole moment, using the result of Prob. 4.2, we can write the charge density of the two-particle configuration as

$$\rho(\mathbf{x}) = -\left(\mathbf{p}\cdot\nabla\right)\left[\frac{1}{r^2}\delta(r-a)\delta(\cos\theta)\left(\delta(\phi-\omega t) - \delta(\phi-\omega t-\pi)\right)\right],$$

and the quadrupole moment can be calculated as

$$\begin{aligned}
Q_{\alpha\beta}(t) &= \int\left(3x_\alpha x_\beta - r^2\delta_{\alpha\beta}\right)\rho(\mathbf{x})d^3x \\
&= \int\frac{1}{r^2}\delta(r-a)\delta(\cos\theta)\left[\delta(\phi-\omega t) - \delta(\phi-\omega t-\pi)\right]\left(p\frac{\partial}{\partial z}\right)\left(3x_\alpha x_\beta - r^2\delta_{\alpha\beta}\right)d^3x.
\end{aligned}$$

It is straightforward to verify that only Q_{xz} and Q_{yz} are non-zero, which are given by

$$\begin{aligned}
Q_{xz}(t) &= \int\frac{1}{r^2}\delta(r-a)\delta(\cos\theta)\left[\delta(\phi-\omega t) - \delta(\phi-\omega t-\pi)\right]3px\,d^3x \\
&= 3pa\left[\cos(\omega t) - \cos(\omega t+\pi)\right] = 6pa\cos(\omega t) \\
&= \mathrm{Re}\left[6pae^{-i\omega t}\right],
\end{aligned}$$

and

$$Q_{yz}(t) = 6pa\sin(\omega t) = \mathrm{Re}\left[6ipae^{-i\omega t}\right].$$

(b) With both magnetic dipole and electric quadrupole moments present, the radiating magnetic field is, as given by Eqs. (9.35) and (9.44),

$$\mathbf{H} = -\frac{k^2}{4\pi}\mathbf{n}\times\left[\mathbf{n}\times\mathbf{m} + \frac{i\omega}{6}\mathbf{Q}(\mathbf{n})\right]\frac{e^{ikr}}{r},$$

where

$$\mathbf{Q}(\mathbf{n}) = \sum_\beta Q_{\alpha\beta}n_\beta = 6pa\begin{pmatrix}0 & 0 & 1\\ 0 & 0 & i\\ 1 & i & 0\end{pmatrix}\begin{pmatrix}\sin\theta\cos\phi\\ \sin\theta\sin\phi\\ \cos\theta\end{pmatrix} = 6pa\begin{pmatrix}\cos\theta\\ i\cos\theta\\ \sin\theta e^{i\phi}\end{pmatrix}.$$

Also,

$$\mathbf{n}\times\mathbf{m} = \omega pa\left(\hat{\mathbf{x}}(i\cos\theta) + \hat{\mathbf{y}}(-\cos\theta) + \hat{\mathbf{z}}\left(-i\sin\theta e^{i\phi}\right)\right).$$

Thus,

$$\mathbf{n}\times\mathbf{m} + \frac{i\omega}{6}\mathbf{Q}(\mathbf{n}) \quad = \quad \omega pa\Bigg[\left(\hat{\mathbf{x}}(i\cos\theta) + \hat{\mathbf{y}}(-\cos\theta) + \hat{\mathbf{z}}\left(-i\sin\theta e^{i\phi}\right)\right)$$

$$
\begin{aligned}
& \quad + \left(\hat{\mathbf{x}}(i\cos\theta) + \hat{\mathbf{y}}(-\cos\theta) + \hat{\mathbf{z}}\left(i\sin\theta e^{i\phi}\right)\right)\Bigg] \\
&= 2\omega pa\big[\hat{\mathbf{x}}(i\cos\theta) + \hat{\mathbf{y}}(-\cos\theta)\big],
\end{aligned}
$$

and

$$
\begin{aligned}
\mathbf{H} &= -\frac{k^2}{4\pi}\mathbf{n} \times \left[\mathbf{n} \times \mathbf{m} + \frac{i\omega}{6}\mathbf{Q}(\mathbf{n})\right]\frac{e^{ikr}}{r} \\
&= -\frac{cpa}{2\pi}k^3\mathbf{n} \times \big[\hat{\mathbf{x}}(i\cos\theta) + \hat{\mathbf{y}}(-\cos\theta)\big]\frac{e^{ikr}}{r} \\
&= -\frac{cpa}{2\pi}k^3\left[(\hat{\mathbf{x}} + i\hat{\mathbf{y}})\cos\theta - \hat{\mathbf{z}}\sin\theta e^{i\phi}\right]\cos\theta\frac{e^{ikr}}{r},
\end{aligned}
$$

which agrees with Jackson up to a sign.

(c) From Eq. (9.21),

$$
\frac{dP}{d\Omega} = \frac{Z_0}{2}r^2|\mathbf{H}|^2 = \frac{Z_0}{2}\frac{c^2p^2a^2}{4\pi^2}k^6(2\cos^2\theta + \sin^2\theta)\cos^2\theta = \frac{1}{8\pi^2\varepsilon_0}ck^6p^2a^2\left(\cos^2\theta + \cos^4\theta\right).
$$

The total time averaged power radiated is

$$
P = \int \frac{dP}{d\Omega}d\Omega = \frac{1}{8\pi^2\varepsilon_0}ck^6p^2a^2 2\pi\int_{-1}^{1}(x^2 + x^4)dx = \frac{1}{4\pi\varepsilon_0}ck^6p^2a^2\frac{16}{15} = \frac{4}{15\pi\varepsilon_0}ck^6p^2a^2.
$$